

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1	2	3	4	5	6	7	8						
A								A					
B								B					
C								C					
D								D					
E								E					
F								F					
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	COVER SHEET MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT WILMAR NABATI INDONESIA & 304	
00	Issued for Approval	22.02.19	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.		18.042-WNI-304.01	Sh.	1
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043		Customer Doc. Nbr.			Cont.	28
1	2	3	4	5	6	7	8						

UniSec

Medium Voltage Switchgear

PT MULTIMAS NABATI ASAHAN

MVD-33 20kV FlourMill

SCHEMATIC DIAGRAM

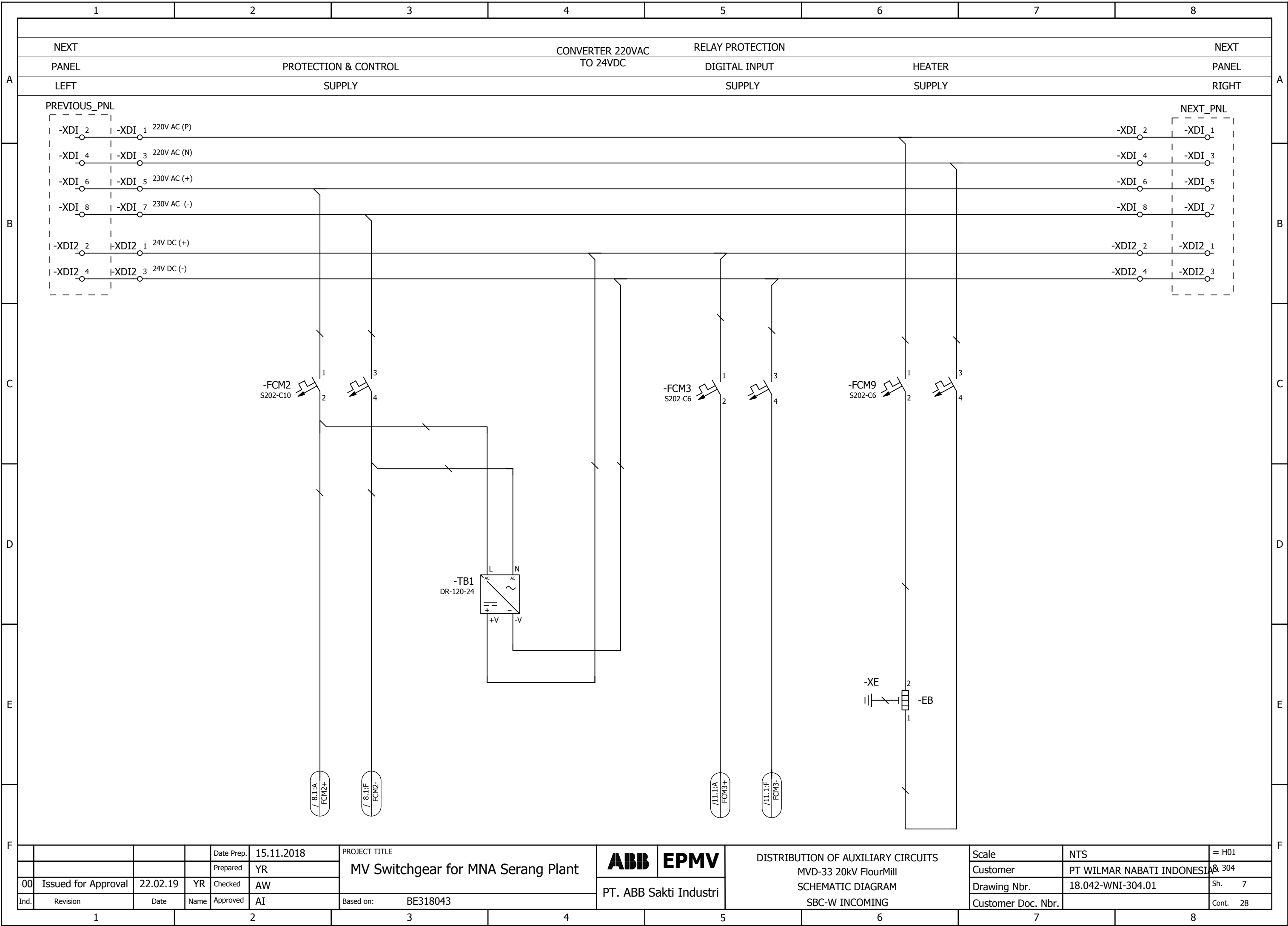
SBC-W INCOMING

H01

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

	1	2	3	4	5	6	7	8						
A	SHEET INDEX								A					
	PAGE			PAGE NAME				REVISION						
	==MVD-33=H01&304/1			COVER SHEET				00						
	==MVD-33=H01&304/2			Index of Sheet				00						
	==MVD-33=H01&304/3			REFERENCE DESIGNATIONS				00						
	==MVD-33=H01&304/4			REFERENCE DESIGNATIONS				00						
	==MVD-33=H01&304/5			REFERENCE DESIGNATIONS				00						
	==MVD-33=H01&304/6			REFERENCE DESIGNATIONS				00						
	==MVD-33=H01&304/7			DISTRIBUTION OF AUXILIARY CIRCUITS				00						
	==MVD-33=H01&304/8			AUXILIARY CIRCUIT				00						
		==MVD-33=H01&304/9			CIRCUIT BREAKER CONTROL				00					
==MVD-33=H01&304/10			AUXILIARY MOTOR & RELAY				00							
==MVD-33=H01&304/11			DIGITAL INPUT/OUPUT RELAY				00							
==MVD-33=H01&304/12			SPARE CONTACT				00							
==MVD-33=H01&304/13			SPARE CONTACT				00							
==MVD-33=H01&304/14			CURRENT TRANSFOMER				00							
		==MVD-33=H01&304/15			VOLTAGE TRANSFORMER				00					
	==MVD-33=H01&304/16			LV DOOR LAYOUT				00						
	==MVD-33=H01&304/17			LV BOX LAYOUT				00						
	==MVD-33=H01&304/18			PART LIST										
	==MVD-33=H01&304/19			Terminal diagram ==MVD-33=H01+-XDA										
	==MVD-33=H01&304/20			Terminal diagram ==MVD-33=H01+-XDB1										
		==MVD-33=H01&304/21			Terminal diagram ==MVD-33=H01+-XDC1									
==MVD-33=H01&304/22			Terminal diagram ==MVD-33=H01+-XDC2											
==MVD-33=H01&304/23			Terminal diagram ==MVD-33=H01+-XDI											
==MVD-33=H01&304/24			Terminal diagram ==MVD-33=H01+-XDI2											
==MVD-33=H01&304/25			Terminal diagram ==MVD-33=H01+-XDI3											
==MVD-33=H01&304/26			Terminal diagram ==MVD-33=H01+-XDI6											
==MVD-33=H01&304/27			Terminal diagram ==MVD-33=H01+-XDV											
	==MVD-33=H01&304/28			Terminal diagram ==MVD-33=H01+-XDX										
E									E					
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant	PT. ABB Sakti Industri	Customer				PT WILMAR NABATI INDONESIA & 304		
00	Issued for Approval	22.02.19	YR	Checked	AW			Based on:	BE318043	Drawing Nbr.		18.042-WNI-304.01	Sh. 2	
Ind.	Revision	Date	Name	Approved	AI			Customer Doc. Nbr.		Cont. 28				
	1	2	3	4	5	6	7	8						

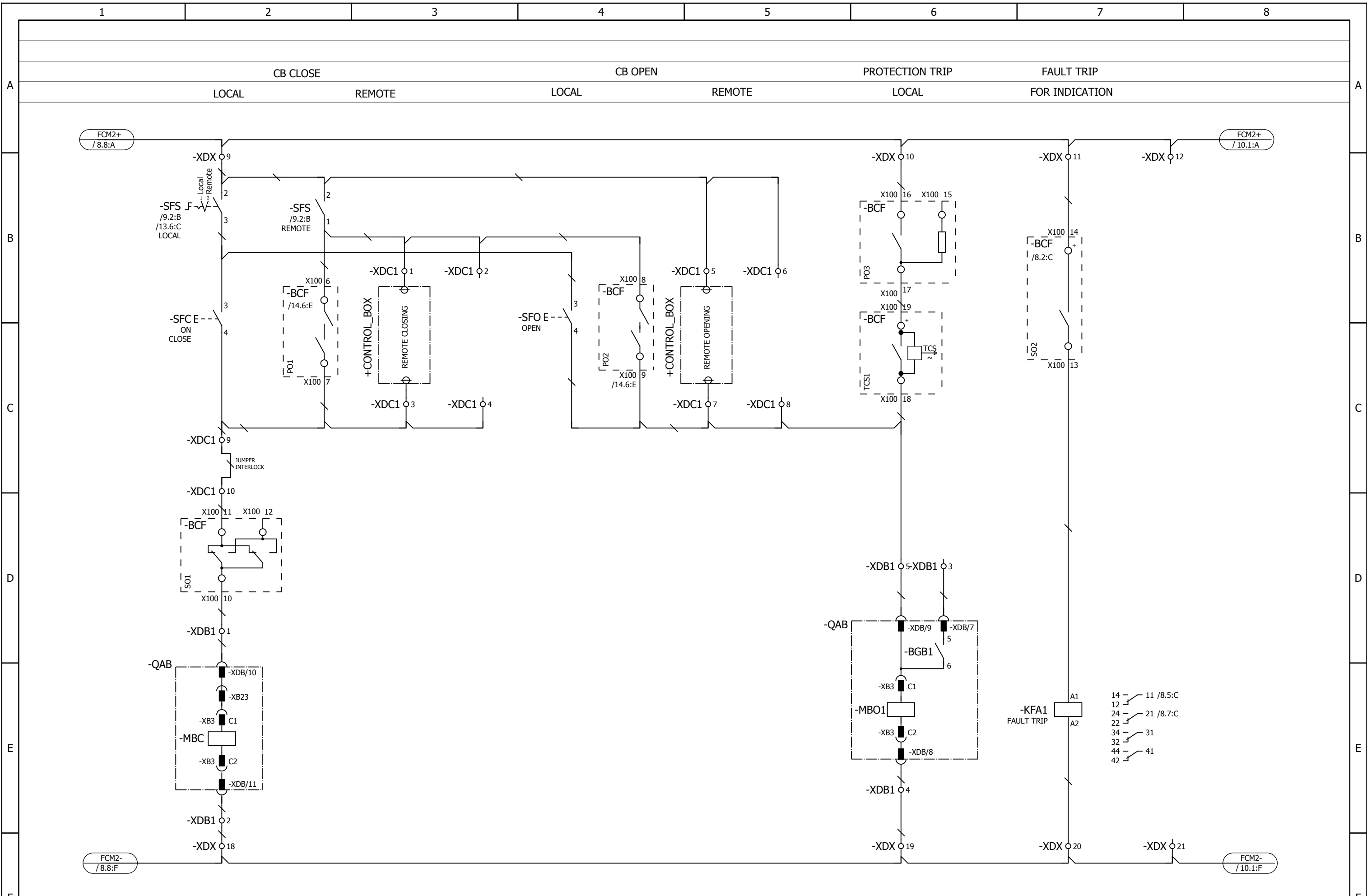
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DISTRIBUTION OF AUXILIARY CIRCUITS MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 304
00	Issued for Approval	22.02.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-WNI-304.01	Sh.	7
Ind.	Revision	Date	Name	Approved	AI	Based on:		BE318043		Customer Doc. Nbr.		Cont.

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY CIRCUIT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR							Customer
00	Issued for Approval	22.02.19	YR	Checked	AW					Drawing Nbr.	18.042-WNI-304.01	Sh. 8
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

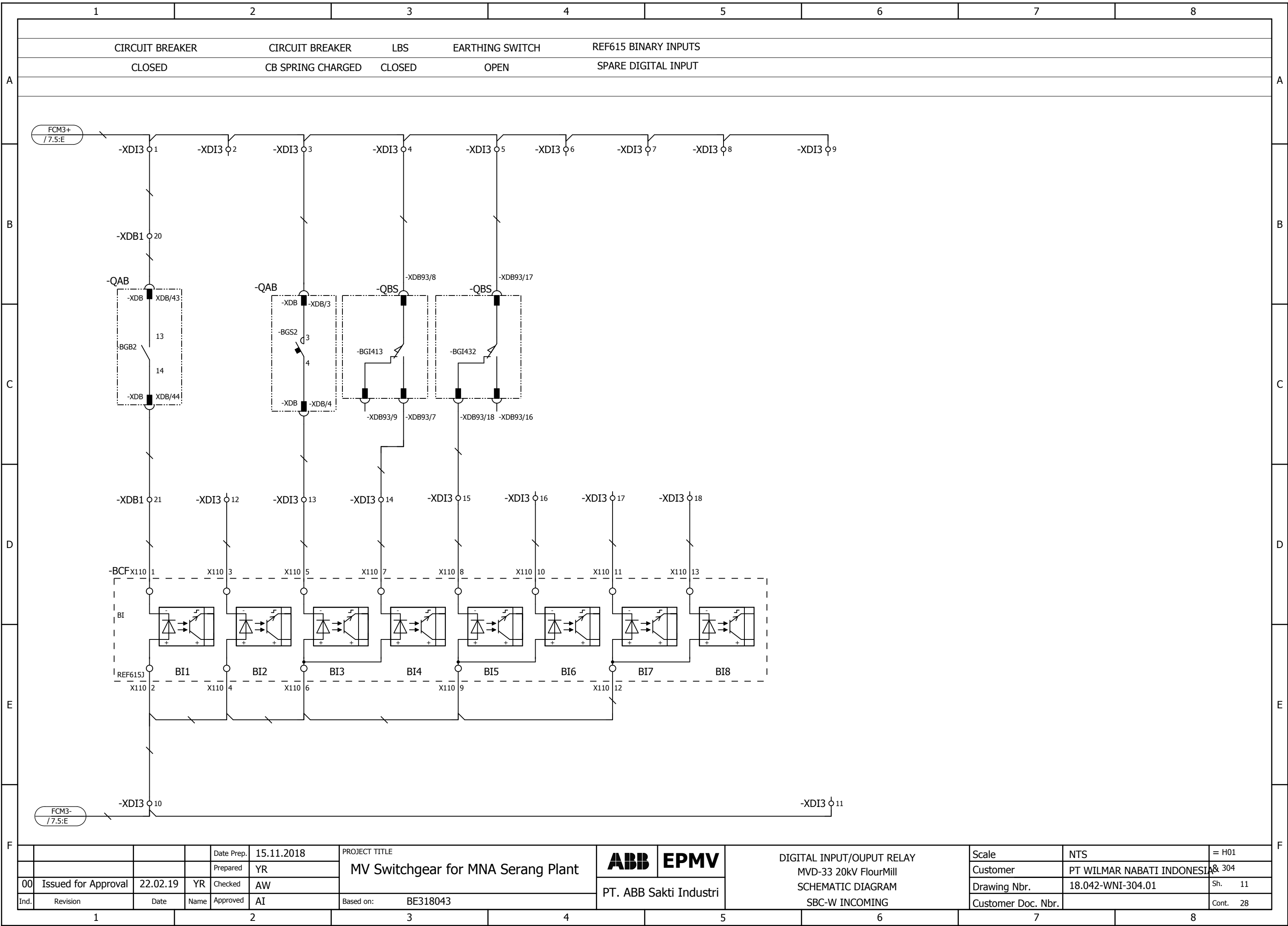
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



		Date Prep.		15.11.2018	PROJECT TITLE		ABB EPMV		CIRCUIT BREAKER CONTROL MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING		Scale	NTS	= H01
		Prepared		YR							Customer	PT WILMAR NABATI INDONESIA	& 304
00 Issued for Approval		22.02.19	YR	Checked							Drawing Nbr.	18.042-WNI-304.01	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043		PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 28

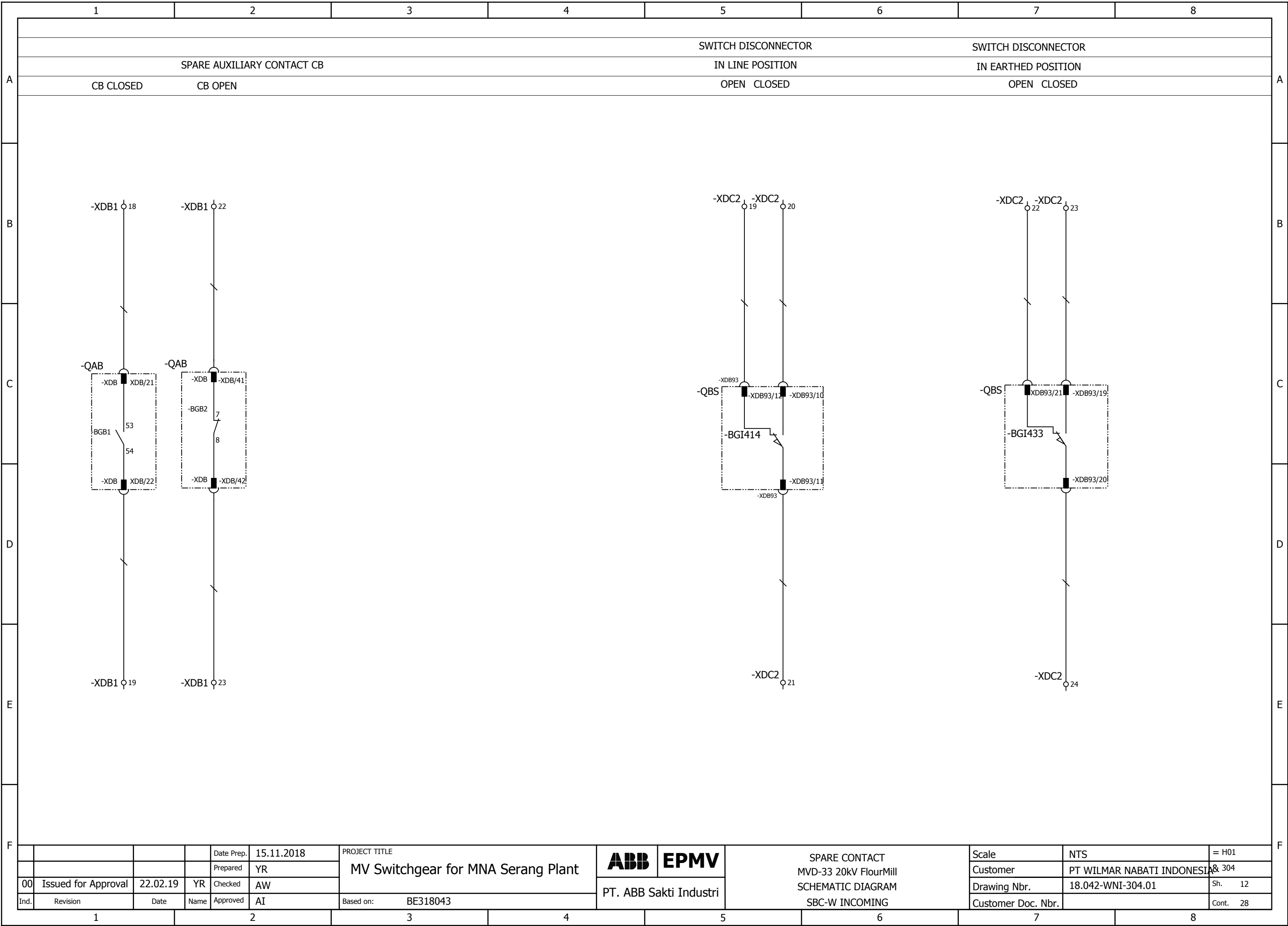
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 304
00	Issued for Approval	22.02.19	YR	Checked	AW		Drawing Nbr.	18.042-WNI-304.01		Sh.	10	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	Customer Doc. Nbr.			Cont.	28
							PT. ABB Sakti Industri					

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



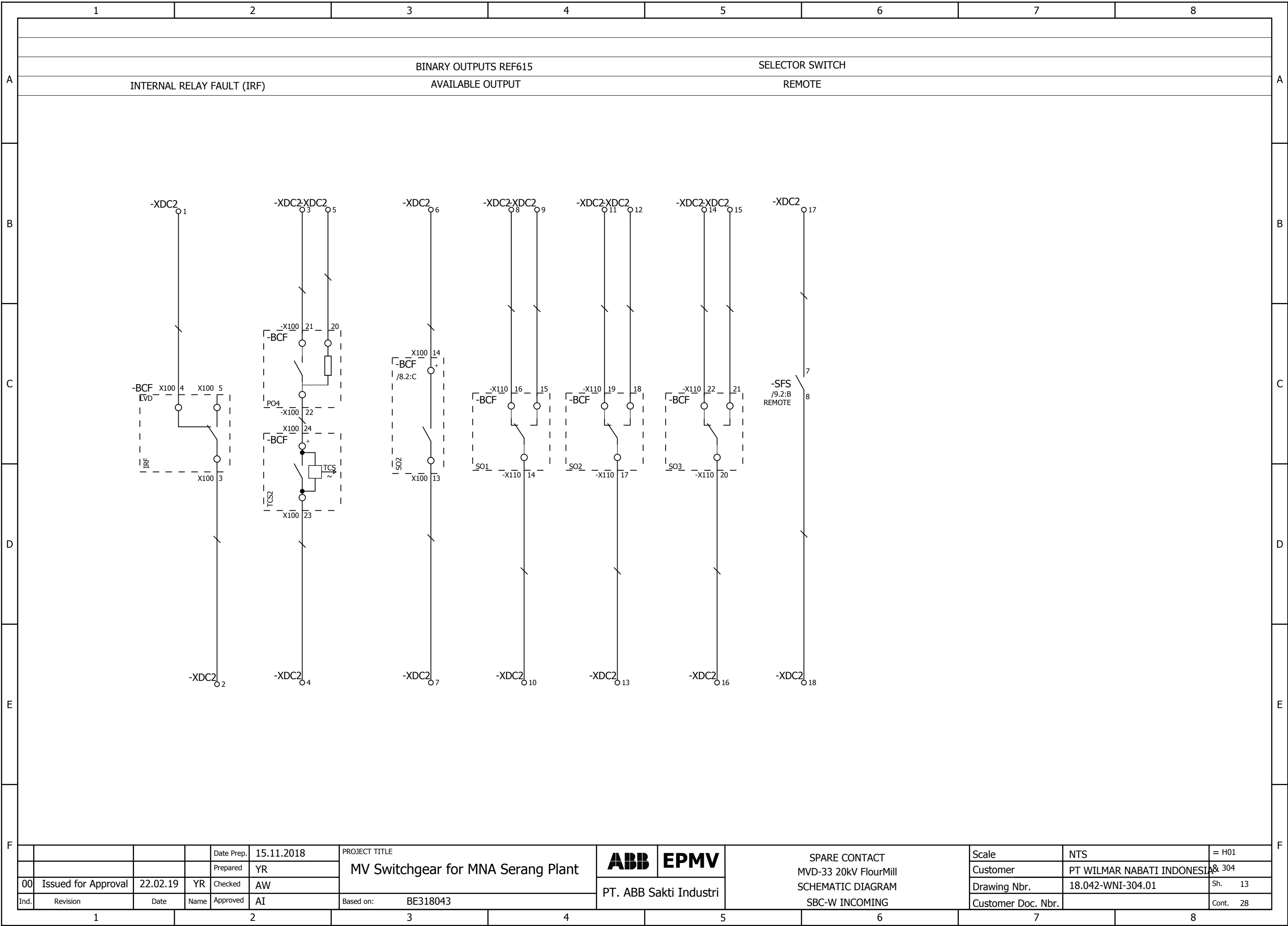
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 304
00	Issued for Approval	22.02.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-WNI-304.01	Sh.	11
Ind.	Revision	Date	Name	Approved	AI	Based on:		BE318043		Customer Doc. Nbr.		Cont.

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

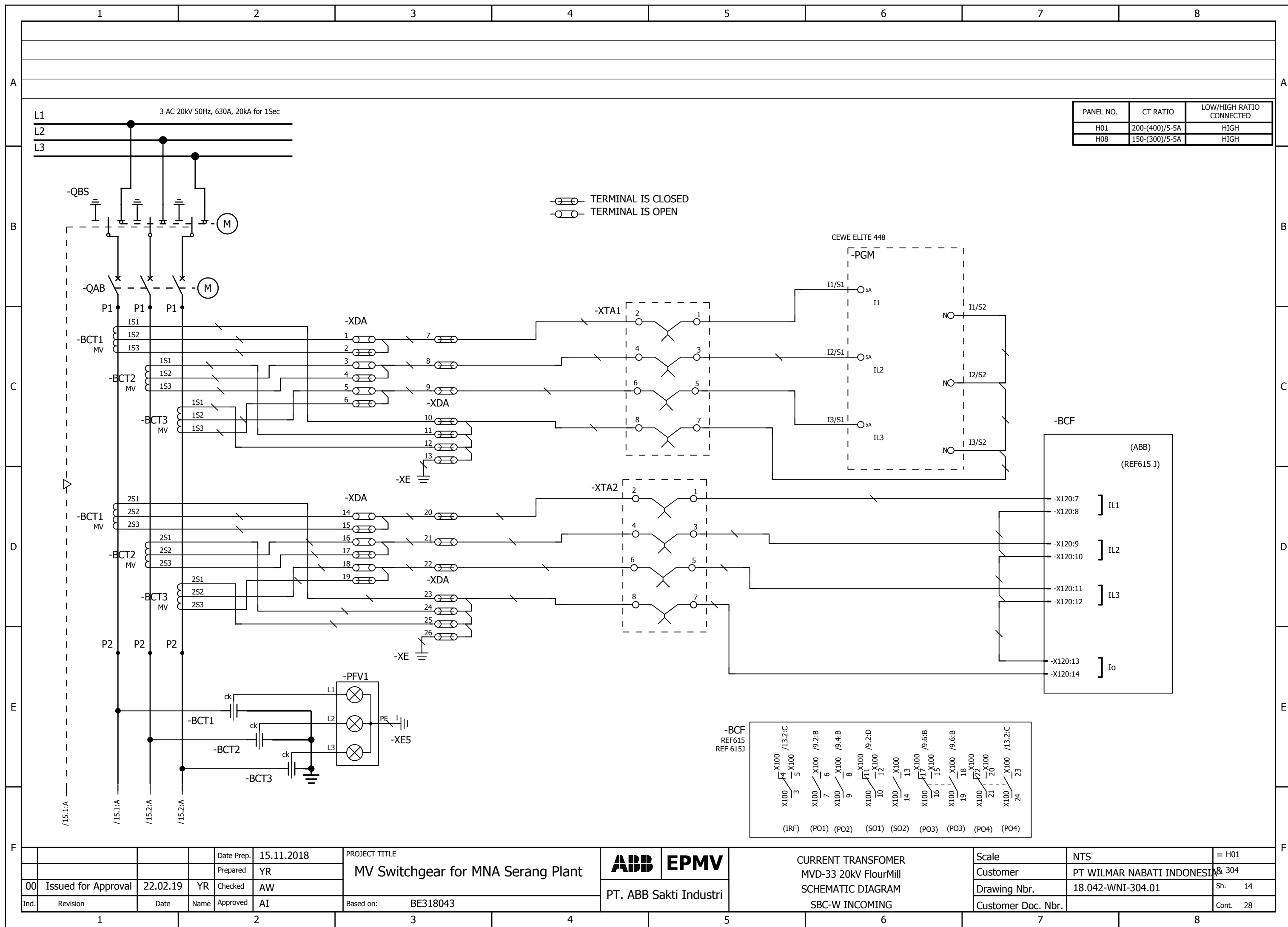


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SPARE CONTACT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR							Customer
00	Issued for Approval	22.02.19	YR	Checked	AW					Drawing Nbr.	18.042-WNI-304.01	Sh. 12
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

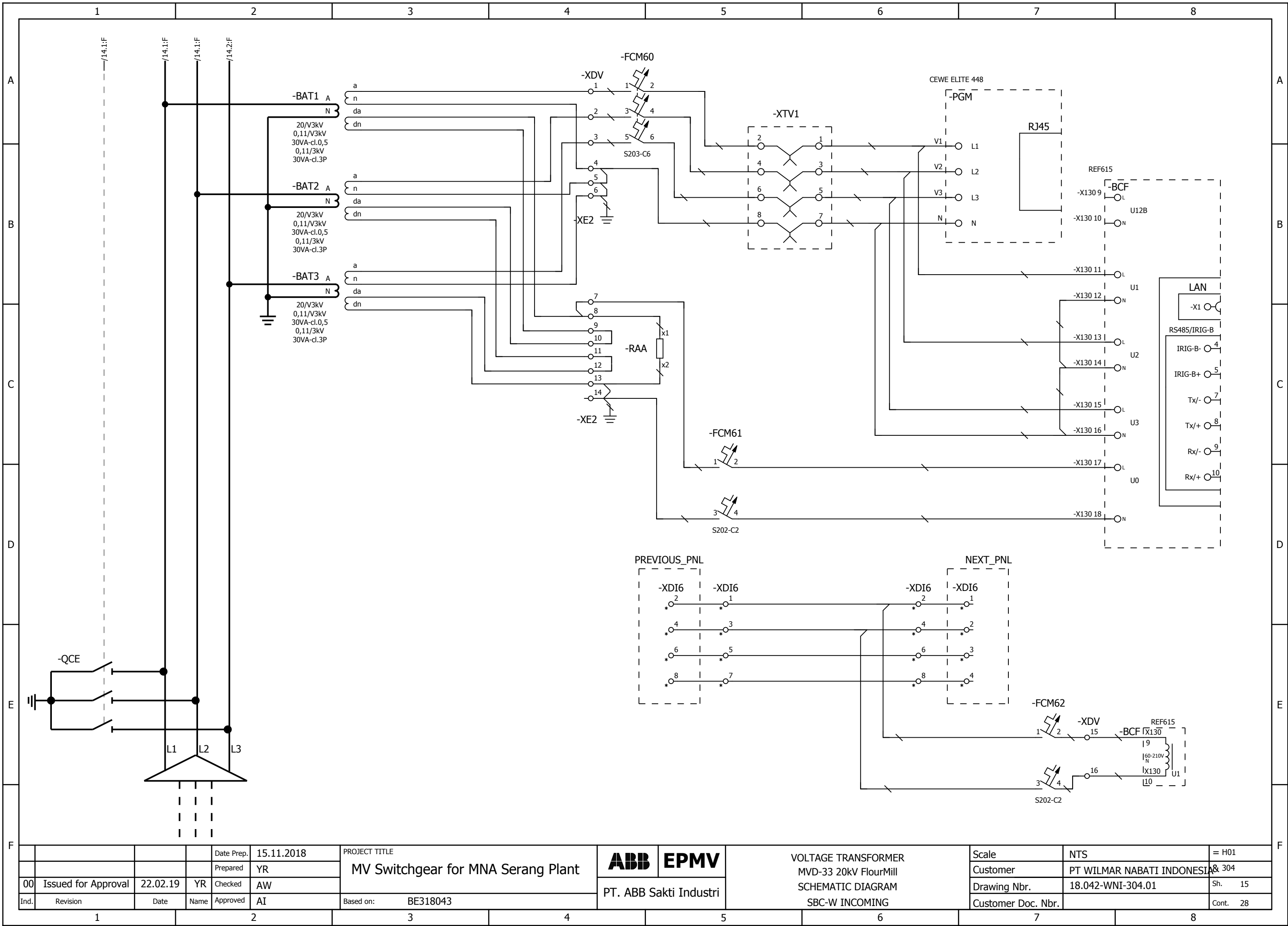
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

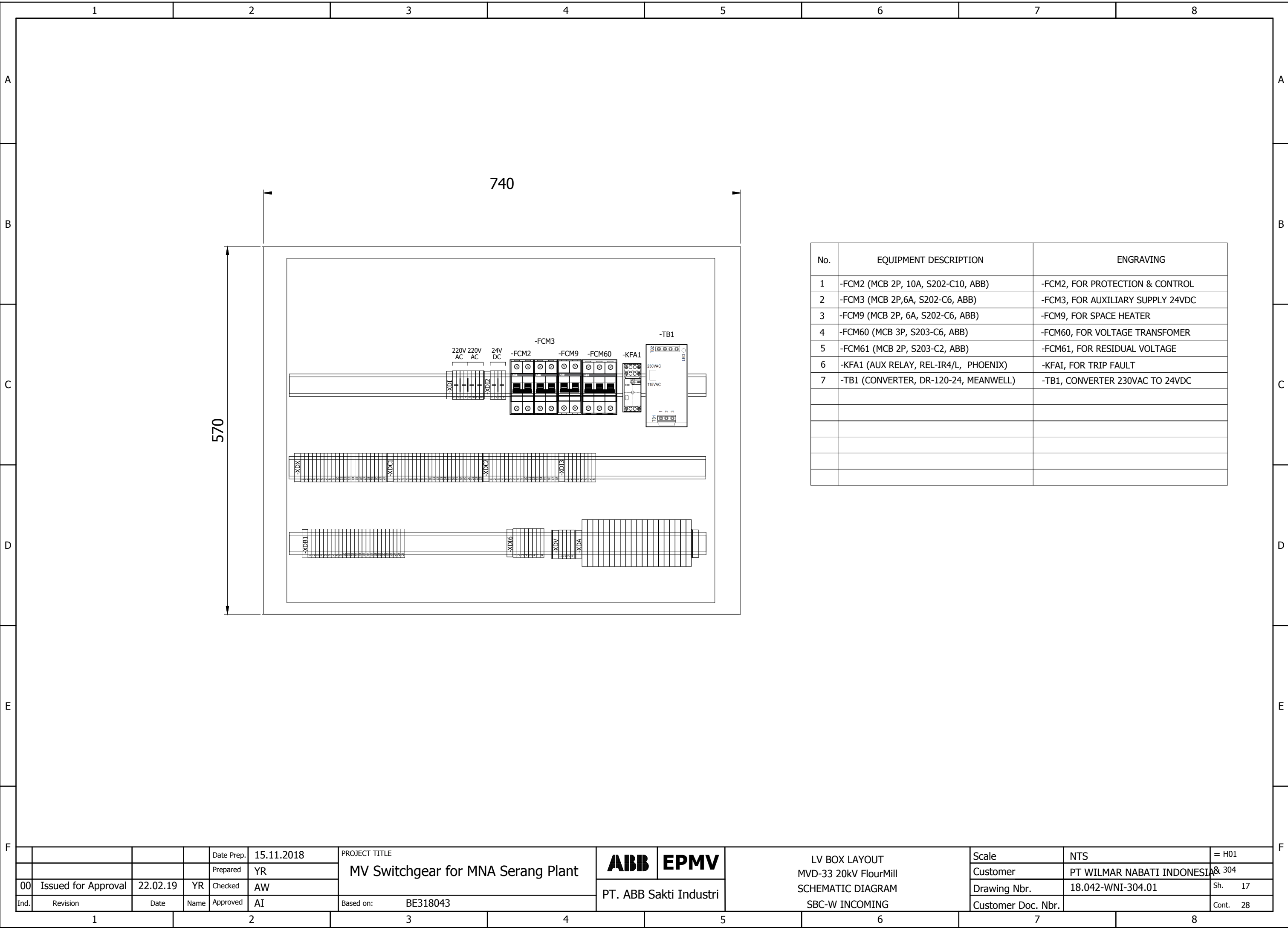


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	PT. ABB Sakti Industri	LV BOX LAYOUT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR				Customer	PT WILMAR NABATI INDONESIA	& 304
00	Issued for Approval	22.02.19	YR	Checked	AW				Drawing Nbr.	18.042-WNI-304.01	Sh. 17
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	Customer Doc. Nbr.		Cont.	28

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8	
Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-BCF		Feeder Management Relay HBFJAEAGABC1ANA11G				ABB		REF615J HBFJAEAGABC1ANA11G		2		/8.2:C;/15.7:B			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.6:E			
-FCM2		MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C			
-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/15.4:A			
-FCM61;-FCM62		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		2		/15.5:D;/15.7:E			
-KFA1		Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.7:E			
-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/8.5:E			
-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/8.4:E			
-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/8.6:E			
-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/8.3:C			
-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1		/9.2:B			
-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1		/9.4:B			
-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1		/9.2:B			
-TB1		Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A				MEANWELL		DR-120-24		1		/7.3:D			
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/14.4:B;/14.4:D			
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/15.5:A			
</															

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

F				E			D		C		B		A	
</														

[illegible]